Label-Level Hierarchical Selective Acceptance of LLM Annotations with Certified Error Control

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Abstract

Large language models (LLMs) routinely annotate items with hierarchical labels, and deploying them means deciding which machine labels may enter a database unreviewed. Existing selective-annotation methods decide per item: an item is accepted with all its labels or reviewed in full. We study the label-level problem and propose HLCL. It scores each predicted node from the agreement of repeated LLM samples and sibling competition, accepts a prefix of each predicted path so that accepted labels are ancestor-closed, and certifies one threshold per depth with a learn-then-test procedure. The certificate bounds the label-level error ratio (wrong accepted labels over accepted labels) with probability $1-\delta$, needs only i.i.d. items, and uses a pooled betting test or an exact depth-wise test chosen on held-out data. Among certified thresholds, HLCL deploys the one with the lowest human effort under an explicit cost model. On four public taxonomies of depth two to four, two LLM annotators and 100 splits per setting, the realised error exceeded the target in at most 8% of splits. Compared with an item-level rule under the same certificate, deciding per label raised the share of correct labels accepted without review from 0.22 to 0.71 on the Blurb Genre Collection at $\alpha = 0.05$. Against the strongest certified baseline in each setting, HLCL had lower mean effort in 14 of 20 settings; after correcting for overlapping splits it is significantly cheaper in 3, significantly more expensive in 1, and not significantly different in the rest.

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Keywords: data quality, data annotation, large language models, selective prediction, risk control, multiple testing, hierarchical classification

1. Introduction

Labelled data is the scarcest input of most knowledge-intensive systems, and the labelling step is where their quality is decided. Human annotation is accurate but slow, and large language models (LLMs) have become a default first-pass annotator: they are cheap, available at scale, and on many taxonomic tasks they agree with human annotators often enough to be useful. Their errors, however, are not uniform. An LLM is typically sure about the coarse category of an item and unsure about the fine one: asking it to tag an abstract as *Medicine* is usually safe, asking it to choose among dozens of medical sub-disciplines is not. A practitioner who deploys such an annotator therefore faces a per-label decision: *which parts of the machine output can be written into the database without review, and which must be confirmed by a person?*

Two constraints shape the answer. The first is a *quality target*: the labels written without review must be right to within a specified error rate α, because they enter downstream aggregations that nobody will audit again. The second is a *reviewing budget*: human attention is the resource being saved, and a procedure that certifies a threshold but routes most of the corpus to reviewers has not solved the problem.

Selective annotation addresses the first constraint. Conformal labeling [1] builds a conformal p-value for each item and applies the Benjamini–Hochberg procedure to control the false discovery rate (FDR) of the items whose machine labels are kept; conformal alignment [2] and selection with conformal p-values [3] give related guarantees, and budget-aware routing [4, 5] addresses the second constraint. These methods share a structural assumption that does not fit hierarchical annotation: the unit of decision is the *item*. An item whose coarse label is certain and whose fine label is a coin flip is either accepted in full, admitting the error, or reviewed in full, paying for work that was only needed one level down. The cost of this granularity can be large. On the Blurb Genre Collection at $\alpha = 0.05$, an item-level rule with exactly our certificate accepts at most 0.22 of the correct labels without review (with the better of its test choices); deciding per label accepts 0.71 (§5.4).

Moving to labels raises two difficulties. The first is statistical. Labels of one item are predicted from the same text and are dependent, and a hierarchy

adds dependence between a node and its descendants; a procedure that treats labels as exchangeable units has no guarantee for the label-level ratio, and reusing calibration data to fit the score that is then tested is known to break selective-risk guarantees [6]. The second difficulty is structural: a label-level rule that ignores the taxonomy may accept a fine label while rejecting its parent, which yields a label set that is not a path and that downstream systems cannot store.

Contributions.

1. We formulate *label-level hierarchical selective acceptance*: for every predicted node, decide accept or review, subject to a high-probability bound on the label-level FDR, with an explicit model of human effort as the objective (§2).
2. We propose HLCL, which combines standard tools for this problem: a node score built from sample agreement and from sibling competition, an ancestor-closed acceptance rule with one threshold per depth, and a learn-then-test calibration in which the candidate thresholds, their order and the test statistic are all fixed on data disjoint from the calibration split (§4). With a single shared threshold, the rule coincides with that of hierarchical selective classification [7]; what changes is the certified quantity.
3. We show that the deployed rule satisfies the label-level bound with probability at least $1 - \delta$ when items are i.i.d., with no assumption on the dependence between labels of the same item. Because the label-level ratio is a weighted average of per-depth rates, an exact depth-wise intersection–union test is available; it remains usable when α is small (§4.3, Appendix A).
4. We evaluate on four public hierarchical corpora of depth two to four with two LLM annotators and 100 random splits per setting, against six baselines, ablations under a fixed test statistic, and sensitivity to the review overhead, the calibration size and δ. Wins and losses are decided by paired confidence intervals corrected for overlapping splits, and we report where HLCL does not help (§5).

2. Problem setting

2.1. Hierarchical labels and annotators

Let $\mathcal{T}$ be a label taxonomy, a rooted tree whose nodes are labels and whose root-to-leaf paths have length at most D. An *item* is a text x_i, and its

gold annotation is a path $(y^\star_{i,1}, \ldots, y^\star_{i,d_i})$ with one node per depth, so every ancestor of a gold node is also gold. This ancestor closure is the convention of hierarchical corpora such as RCV1 [8], Web of Science and the Blurb Genre Collection.

An annotator maps x_i to a predicted path $(\hat{y}_{i,1}, \ldots, \hat{y}_{i,\hat{d}_i})$. In this paper it is an LLM queried k times with temperature sampling, but the method only needs a score for every predicted node. A predicted node at depth d is *correct* if the prediction agrees with the gold path up to depth d; a wrong fine label is therefore an error even when the coarse label is right.

A decision rule with parameters λ accepts some predicted nodes. Let $A(\lambda)$ and $W(\lambda)$ be the numbers of accepted and of accepted-and-wrong nodes of a random item. We define the label-level false discovery rate as the ratio of expectations (a marginal FDR, *mFDR*)

$$\mathrm{FDR}(\lambda) \;=\; \frac{\mathbb{E}[W(\lambda)]}{\mathbb{E}[A(\lambda)]}, \tag{1}$$

the long-run share of wrong labels among the labels written without review, with the convention $\mathrm{FDR}(\lambda) = 0$ when $\mathbb{E}[A(\lambda)] = 0$ (a rule that accepts nothing makes no error). The node score is fitted on separate data (§4.1) and is treated as fixed, so (1) is defined conditionally on the fitted score. This is the quantity an operator monitors, and it does not require labels of the same item to be independent.

Form of the guarantee. The rule is chosen from a labelled calibration sample, so $\mathrm{FDR}(\hat{\lambda})$ is a random quantity. We ask that

$$\Pr\big(\mathrm{FDR}(\hat{\lambda}) \le \alpha\big) \;\ge\; 1 - \delta, \tag{2}$$

with the probability taken over the calibration sample. Conformal labeling [1] instead controls the expected false discovery proportion $\mathbb{E}[\mathrm{FDP}]$ over calibration and test draws. The two guarantees bound different functionals and neither implies the other: ours fixes the target functional (the population ratio (1)) and guarantees that the *deployed* rule meets it with probability $1-\delta$; theirs averages the realised proportion over repetitions. The high-probability form suits an operator who must vouch for one released dataset; we compare the two empirically in §5.4.

2.2. Human effort

Reviewing an item does not have a fixed cost: a reviewer who is told that the coarse label is certified and only chooses the sub-category does less work than one who annotates the item from scratch. Let $A_{i,d} \in \{0, 1\}$ indicate that the depth-d node of item i was accepted. The cost of item i is

$$c_i \;=\; \sum_{d=1}^{D} \omega_d \, \mathbf{1}\{A_{i,d} = 0\} \;+\; c_0 \, \mathbf{1}\{\exists d : A_{i,d} = 0\}, \tag{3}$$

where $\omega_d \geq 0$ with $\sum_d \omega_d = 1$ is the share of a full annotation spent on level d, and c_0 is a fixed overhead for opening and reading an item. A fully accepted item costs 0 and a fully manual one costs $1 + c_0$. We report effort as $\mathbb{E}[c]/(1 + c_0)$, the fraction of the cost of annotating the corpus by hand. The overhead matters: without it, the cheapest rule would touch nearly every item to fix only its fine level. We use $c_0 = 0.5$ and equal ω_d and vary c_0 in §5.6. Neither value comes from a timing study; they are a transparent default, and the sensitivity analysis shows that the guarantee does not depend on them and the ranking of methods changes little.

2.3. Objective

For a risk level α and a confidence $1 - \delta$ we seek

$$\hat{\lambda} \in \arg\min_{\lambda} \; \mathbb{E}[c(\lambda)] \quad \text{subject to (2)}. \tag{4}$$

The decision is made per node, so a rule may accept the coarse level of an item and route only its finer levels. The constraint is on the pooled label-level ratio, which is weaker, and hence easier to meet, than bounding every depth separately.

Definition 1 (Ancestor closure)**.** A decision rule is *ancestor-closed* if it accepts a node only when it accepts all predicted ancestors of that node.

Closure is required for the output to be usable. A label set that contains a fine label without its parent is not a path in $\mathcal{T}$: it cannot be stored in a path-keyed table and fails any taxonomy validator. Closure also matches the task: if the annotator hesitates between *Cardiology* and *Neurology*, the useful partial output is the certified parent *Medicine*.

3. Related work

Selective annotation with guarantees. Huang et al. [1] propose conformal labeling: a conformal p-value per item, computed against calibration items that the model mislabelled, followed by Benjamini–Hochberg at level α, controls the expected FDR of the set of items whose machine labels are used. Gui et al. [2] apply the same construction to decide when the output of a foundation model can be trusted, and Jin and Candès [3] develop the underlying theory of selection with conformal p-values. All three select whole items and assume one label per item. HLCL differs in the unit of decision, in enforcing ancestor closure and in the form of the guarantee.

Routing between humans and models. Jung et al. [4] decide when an LLM judge should defer to a human, using learn-then-test to bound disagreement with human judgments. Purificato et al. [5] route instances to human or LLM annotators under a budget using difficulty features, without a statistical guarantee. Both decide per item and address how review capacity is spent; they can be combined with HLCL, which decides what needs review in the first place.

Selective and hierarchical selective classification. Selective classification lets a classifier abstain; Geifman and El-Yaniv [9] give a threshold rule with a high-probability bound on the risk of the accepted predictions, the direct ancestor of our item-level certified baseline. Goren et al. [7] extend this to hierarchies: instead of abstaining, the classifier backs off to an ancestor of its prediction, and a threshold is calibrated so that the returned nodes meet a target accuracy with high probability. This is the closest work to ours and our ancestor-closed rule has the same shape. The differences are in the target and the decision. Hierarchical selective classification bounds the error of the returned node over all items, including items that back off to the root, and chooses the threshold that returns the most specific nodes; we bound the error among the labels written without review (a false discovery rate), certify a separate threshold per depth with two complementary tests, and choose among certified thresholds by human effort. We include an adaptation of it as a baseline (§5.2). Surveys of hierarchical classification [10] describe the top-down, per-level decision structure that our rule inherits.

Guarantees below the item. Mohri and Hashimoto [11] remove claims from a generated text until the remainder is certified factual, a back-off close in

spirit to our ancestor closure; the unit is a claim rather than a taxonomy node, and there is no budget. Gurram [6] controls the selective risk per field in document extraction, the label-level granularity we adopt, and documents how score refitting and clustering of fields within a document invalidate naive procedures. Fields of a form are not organised in a hierarchy, and that work does not accept a coarse unit while routing a fine one; we use its field-level procedure as a baseline. Hierarchical conformal classification [12, 13] and tree-based conformal prediction for multi-label data [14] return mixed-level prediction sets with coverage guarantees. Coverage asks that the truth lie in the returned set; we ask that the labels written without review be right, which is a different requirement, and none of these methods models human effort.

LLM annotations in statistical inference. Prediction-powered inference [15] and its use with LLM annotations [16] combine many machine labels with a few human labels to obtain valid confidence intervals for population quantities. Their goal is an estimate, not a per-item decision about which labels to keep; they are complementary to HLCL, which decides which labels can be stored.

Calibration. Post-hoc calibration such as temperature scaling [17] improves the reliability of confidence scores. A calibrated score still leaves open where to cut, and the cut must be justified on labelled data; HLCL does not need its score to be calibrated for the guarantee to hold.

Learn-then-test and betting tests. Our calibration is an instance of learn-then-test [18]: candidate parameters are hypotheses tested in a fixed sequence [19], which controls the family-wise error rate without splitting the error budget across candidates. We use the betting p-values of Waudby-Smith and Ramdas [20]. Conformal risk control [21] bounds an expected monotone loss and could replace our certificate when an expectation guarantee suffices. The individual statistical tools we use are standard. What is specific to our setting is the hypothesis family (per-depth thresholds under ancestor closure), the observation that the label-level ratio is a weighted average of per-depth rates, which makes an exact intersection–union test available, the selection between the two tests on held-out data, and the effort-based choice among certified thresholds. We regard the formulation and its evaluation, rather than any single tool, as the contribution.

4. Method

HLCL has four parts: a node score (§4.1), an ancestor-closed acceptance rule (§4.2), a certificate (§4.3) and the choice of the deployed threshold (§4.4). Algorithm 1 summarises the procedure.

4.1. Node score from repeated annotation

The annotator is queried k times with temperature sampling and asked for one path and a stated confidence per level. At each depth, the k paths that agree with the prefix chosen so far define a distribution over candidate nodes, and the *consensus path* follows its mode. A node v at depth d is described by

- **vote**: the share of the k samples whose path agrees with the consensus up to v; this is non-increasing along a path;
- **conditional agreement**: $\mathrm{vote}(v)/\mathrm{vote}(\mathrm{parent}(v))$;
- **sibling margin and entropy**: the gap between the two largest conditional shares at depth d and the entropy of the conditional distribution;
- **stated confidence**: mean and standard deviation of the confidence the annotator reports for this level over the supporting samples;
- the parent's vote, the logarithm of the parent's fan-out in $\mathcal{T}$, and a depth indicator.

A logistic regression on these features, fitted on a *fit split* that is disjoint from the calibration and test data, maps each node to a score $s \in (0, 1)$. Unlike the vote, s is not monotone along a path, because a child can face weaker competition than its parent. Ancestor closure is therefore a real constraint on the rule. Keeping the fit split separate prevents the score-refit leakage described in [6].

4.2. Acceptance rule

Given thresholds $\lambda = (\lambda_1, \ldots, \lambda_D)$, one per depth, the rule accepts the depth-d node of item i when

$$A_{i,d} \;=\; \prod_{e \le d} \mathbf{1}\{s_{i,e} \ge \lambda_e\} = 1, \tag{5}$$

where $s_{i,e}$ is the score of the node of item i at depth e. The accepted set of every item is a prefix of its predicted path and hence ancestor-closed (Appendix A). A reviewer of a partially accepted item receives the certified prefix and decides only the remaining levels, which is the cost model (3).

We use one threshold per depth rather than one per node. A per-node rule has as many parameters as $\mathcal{T}$ has nodes and cannot be certified from a calibration sample of realistic size; a per-depth rule has D parameters and is unaffected by adding leaves at an existing depth. The candidate set contains the grid of per-depth threshold vectors built from score quantiles on the fit split and, as a special case, the vectors with equal thresholds at all depths.

4.3. Certificate

For item i let $A_i(\lambda) = \sum_d A_{i,d}$ and $W_i(\lambda) = \sum_d A_{i,d}\mathbf{1}\{C_{i,d} = 0\}$, where $C_{i,d}$ indicates that the node is correct, and define the loss

$$L_i(\lambda) \;=\; W_i(\lambda) - \alpha A_i(\lambda) \;\in\; [-\alpha D,\, (1-\alpha)D]. \tag{6}$$

If items are i.i.d., so are $L_1(\lambda), \ldots, L_n(\lambda)$, and

$$\mathbb{E}[L(\lambda)] \le 0 \iff \mathbb{E}[W(\lambda)] \le \alpha\, \mathbb{E}[A(\lambda)] \iff \mathrm{FDR}(\lambda) \le \alpha \tag{7}$$

whenever $\mathbb{E}[A(\lambda)] > 0$. A valid p-value for the null $H_\lambda : \mathrm{FDR}(\lambda) > \alpha$ is therefore all that is needed. The labels of one item may be arbitrarily dependent, because each item contributes one scalar L_i.

Pooled betting test. Rescale L to $x \in [0,1]$ and let t be the rescaled zero. Under H_λ, $\mathbb{E}[x] > t$, and the capital $K_m = \prod_{i \le m}\big(1 + \ell_i(t - x_i)\big)$ with predictable bets $\ell_i \in [0,\, 0.5/(1-t)]$ is a nonnegative supermartingale, so $p^{\mathrm{bet}} = 1/\max_m K_m$ is a valid p-value by Ville's inequality [20]. We use the predictable plug-in bet $\ell_i = \min\{\sqrt{2\log(20)/(n\hat{\sigma}^2_{i-1})},\, 0.5/(1-t)\}$, where $\hat{\sigma}^2_{i-1}$ is a regularised running variance of $x_1, \ldots, x_{i-1}$ and the items are visited in a fixed random order; the bet adapts to the small variance of L at conservative thresholds. Any predictable bet in the stated range gives a valid p-value.

Depth-wise intersection–union test. Let $q_d = \mathbb{E}[W_d]/\,\mathbb{E}[A_d]$ be the error rate among nodes accepted at depth d. The pooled rate is a weighted average of these,

$$\mathrm{FDR}(\lambda) = \frac{\sum_d \mathbb{E}[W_d]}{\sum_d \mathbb{E}[A_d]} = \sum_d \frac{\mathbb{E}[A_d]}{\sum_e \mathbb{E}[A_e]}\, q_d, \tag{8}$$

so $\mathrm{FDR}(\lambda) > \alpha$ implies $q_d > \alpha$ for some d, i.e. $H_\lambda \subseteq \bigcup_d H_d$ with $H_d : q_d > \alpha$. Because the pairs $(A_{i,d}, W_{i,d})$ are i.i.d. across items, conditionally on which items are accepted at depth d their error indicators are independent with success probability q_d. With n_d acceptances and w_d errors,

$$p_d = \Pr\big(\mathrm{Bin}(n_d, \alpha) \le w_d\big) \tag{9}$$

is an exact p-value for H_d; if a depth has a finite threshold but no acceptance in the sample we set $p_d = 1$, and depths with an infinite threshold are omitted. By the intersection–union principle,

$$p^{\mathrm{iut}} = \max_d p_d \tag{10}$$

is a valid p-value for H_λ without multiplicity correction (Appendix A). The test depends on the score only through which nodes are accepted. It is coarser than the pooled test, since it asks every depth to meet α, but it uses exact tails and remains usable when αn is small.

Choosing the test. Neither test dominates the other: the pooled test exploits the constraint that is actually the target and is sharper when depths differ in error rate, while the depth-wise test is the only one that certifies anything when α is very small. HLCL uses the pooled test if, on the fit split, the sequence it induces certifies at least one candidate, and the depth-wise test otherwise. The rule depends only on the fit split, so it does not affect the guarantee. The fit-split scores used for these decisions are in-sample (the score was trained on the same items), which makes the rule slightly optimistic; we tried two-fold cross-fitting instead and found it made the candidate order worse, not better, because each half-model is trained on half as many items (in a pilot on DBPedia, effort at $\alpha = 0.1$ rose from 0.00 to 0.06). We therefore keep the in-sample variant; either choice is valid.

Theorem 1 (Validity)**.** *Assume the fit, calibration and test items are i.i.d. Let the candidate set, their order and the choice of test be measurable functions of the fit split, and let candidates be tested on the calibration split in that order, stopping at the first p-value above δ. Then, with probability at least $1-\delta$ over the calibration split and conditionally on the fit split, every certified candidate λ satisfies $\mathrm{FDR}(\lambda) \le \alpha$. In particular the deployed $\hat{\lambda}$ satisfies* (2)*.*

The proof is in Appendix A. The guarantee is about the deployed rule: any selection among certified candidates, including the effort criterion below, preserves it, because all of them are covered simultaneously.

Algorithm 1 HLCL

Require: items, annotator, taxonomy $\mathcal{T}$, risk α, confidence δ, cost model (ω, c_0)

query the annotator k times per item; build consensus paths and node features
split the labelled items into fit, calibration and test parts
fit the node score s on the fit split
build candidate threshold vectors from score quantiles on the fit split
choose the test statistic on the fit split (§4.3)
order candidates by their p-value on the fit split, conservative first among ties
for each candidate λ in this order **do**
    compute its p-value on the calibration split
    if $p > \delta$ **then break**
    end if
    record λ and its estimated effort $\hat{c}(\lambda)$
end for
return the recorded λ with the smallest $\hat{c}$ (accept nothing if none)

4.4. Choosing the deployed threshold

Certification leaves a set of admissible thresholds. Selective annotation methods usually deploy the most permissive one, which maximises the number of accepted labels. HLCL instead estimates the effort (3) of each certified candidate on the calibration split and deploys the cheapest. The two criteria differ when a threshold accepts slightly more labels but leaves more items partially open, each of which pays the overhead c_0.

5. Experiments

5.1. Data and annotators

We use four public hierarchical corpora whose fine levels differ in difficulty: DBPedia classes (9/70/219 labels at three levels), WOS-46985 (7/134 labels at two levels), the Blurb Genre Collection (three levels, 147-edge official hierarchy; we keep the blurbs whose genre path is unique and has at least three levels) and the Home & Kitchen product categories of Amazon Reviews 2023 [22] (four levels, 8/31/112/296 nodes; from the first 400,000 records of the metadata file we keep products whose category path has at least four

Table 1: Accuracy of the consensus path by depth.

Dataset (annotator)	level 1	level 2	level 3	level 4
DBPedia (A)	0.987	0.910	0.880	—
WOS-46985 (A)	0.749	0.470	—	—
Blurb (A)	0.978	0.853	0.710	—
Blurb (B)	0.891	0.735	0.589	—
Amazon (A)	0.969	0.899	0.828	0.731

levels below the root and paths with at least 100 products, a pool of 154,787 products, and use the title, the first 500 characters of the features and the first 400 of the description, truncated to 1,200 characters, as text). From each we draw 3,000 items with a fixed seed. Annotator A (`deepseek-chat`) labels all four corpora; annotator B (`claude-haiku-4-5`), from a different model family, labels the Blurb collection again. Each item is queried $k = 5$ times at temperature 1.0. All 75,000 responses are cached and released, so every number below can be reproduced without API access. Four responses could not be parsed (three blank ones on WOS-46985 and one truncated one from annotator B) and count as missing samples; on Amazon, three responses left every level empty and count as votes for no label.

Table 1 gives the per-depth accuracy of the consensus path. DBPedia is easy at every level; the Blurb collection degrades at the third level, more for annotator B; Amazon degrades steadily with depth, from 0.97 at the first level to 0.73 at the fourth; WOS-46985 is hard at the second level, where the annotator is right less than half the time. In that regime no certified method can accept much at small α, and we report it as such.

5.2. Baselines

All methods use the same learned node score s and the same splits.

- **Item-level certified selection** (Item-cert.): each item is scored by its weakest node and one threshold is certified, so the item is accepted or reviewed as a whole. This is the high-probability selective rule of [9] applied to items, with our certificate. We run it with the automatic test choice of HLCL, with the pooled test only and with the depth-wise test only, and in every table report the variant with the lowest mean effort in that setting (Item-cert.*). Choosing this variant on the test results is an oracle choice that can only favour the baseline.

- **Hierarchical selective classification** (HSC) [7]: each item returns the deepest prefix of its predicted path whose nodes all clear one shared threshold. The original method scores a node by the aggregated probability of its descendants under a flat classifier; we use our node score instead, so that all methods share one score. Following that work, the risk is the item-level error over all items (an item backed off to the root counts as correct), certified with an exact binomial tail and fixed sequence testing, and the most permissive certified threshold is deployed. Its guarantee is thus on a different quantity from ours; we report it because it is the closest published hierarchical back-off method.

- **Per-field selective risk control** (Per-field) [6]: one threshold per depth, certified with an exact binomial tail at level δ/D (Bonferroni over depths), as in the field-i.i.d. tier of that work. Candidate thresholds and their order come from the fit split; the most permissive certified threshold is deployed. Depths are treated as independent fields and closure is not enforced.

- **Item-level conformal selection** (Item-conf.) [1, 3]: a conformal p-value per item from the item score, followed by Benjamini–Hochberg at level α. It controls the *expected* item-level FDR, which upper-bounds the expected label-level FDR, but does not give (2). It does not use the fit split.

- **Depth-wise Benjamini–Hochberg** (Depth-BH): conformal p-values per node and Benjamini–Hochberg separately at each depth. Nodes of one depth are i.i.d. across items, so this controls the expected FDP at each depth; it gives no high-probability statement and no direct statement about the pooled ratio.

- **Fixed threshold** ($s \geq 0.9$): the common practitioner rule, with no guarantee.

Item-cert., HSC and Per-field carry high-probability certificates; we call them the *certified baselines*. Item-conf. and Depth-BH carry expected-value guarantees.

5.3. Protocol

For each corpus, annotator and α we draw 100 random splits into fit, calibration and test parts (600/1200/1200 items). The score, the candidates,

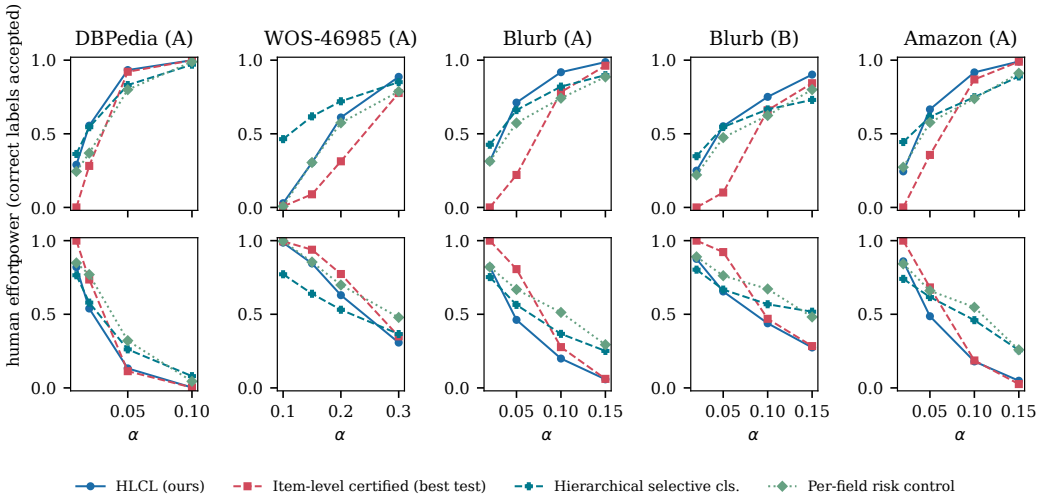


Figure 1: Power (top) and effort (bottom) against α for HLCL and the three certified baselines.

their order and the test statistic are fixed on the fit split, certification uses the calibration split only, and all reported numbers are computed on the test split with $\delta = 0.1$ unless stated otherwise. We report (i) *power*, the share of correct label nodes accepted without review; (ii) *effort* (3) as a fraction of fully manual annotation; and (iii) *viol.*, the share of splits whose realised label-level error on the test split exceeds α. The guarantee concerns the population rate, so viol. also counts test-sample fluctuations and is a conservative check: a valid method should stay near or below $\delta = 0.1$. The values of α differ by corpus because a useful range depends on the annotator's accuracy.

To compare two methods in a setting we use the paired difference of their effort over the 100 splits. The splits are drawn from the same 3,000 items, so the differences are positively correlated and the naive standard error is too small; we therefore use the corrected resampled-t variance of Nadeau and Bengio [23], which multiplies the sample variance by $1/J + n_{\text{test}}/n_{\text{train}} = 0.677$ (with $J = 100$ splits, $n_{\text{test}} = 1200$, $n_{\text{train}} = 1800$). HLCL *wins* a setting if the corrected 95% interval for (HLCL − baseline) lies below zero, *loses* if it lies above zero, and *ties* otherwise.

5.4. Main results

Table 2 and Figures 1–2 summarise the results.

Table 2: Effort / power (100 splits). Lower effort is better. The first four columns carry high-probability certificates; Item-conf. and Depth-BH control an expectation; $s \geq 0.9$ has no guarantee. Item-cert.* is the best of three test choices in each row. Bold: lowest effort among the certified methods that met the label-level target in the row. †: the realised error exceeded α in more than 10% of splits.

Setting	α	HLCL	Item-cert.*	HSC	Per-field	Item-conf.	Depth-BH	$s \geq 0.9$
DBPedia (A)	0.01	0.82/0.29	1.00/0.00	**0.77/0.36**	0.85/0.24	0.83/0.19†	0.69/0.47†	0.17/0.91†
	0.02	**0.54/0.56**	0.74/0.28	0.58/0.54	0.77/0.37	0.51/0.52	0.53/0.62†	0.17/0.91†
	0.05	0.13/0.93	**0.11/0.92**	0.26/0.83	0.32/0.80	0.18/0.86	0.14/0.93	0.17/0.91
	0.1	0.00/1.00	**0.00/1.00**	0.08/0.97	0.04/0.99	0.03/0.99	0.02/1.00	0.17/0.91
WOS-46985 (A)	0.1	**0.99/0.03**	1.00/0.01	0.77/0.46†	1.00/0.01	0.94/0.09†	0.84/0.32†	0.97/0.06†
	0.15	**0.85/0.30**	0.94/0.09	0.64/0.62†	0.85/0.30	0.88/0.18	0.69/0.57†	0.97/0.06
	0.2	**0.63/0.61**	0.77/0.31	0.53/0.72†	0.70/0.57	0.80/0.28	0.57/0.71†	0.97/0.06
	0.3	**0.31/0.89**	0.35/0.78	0.37/0.85	0.48/0.79	0.59/0.53	0.39/0.85	0.97/0.06
Blurb (A)	0.02	0.82/0.31	1.00/0.00	**0.75/0.43**	0.82/0.31	0.99/0.01	0.69/0.53†	0.51/0.71†
	0.05	**0.46/0.71**	0.81/0.22	0.56/0.66	0.67/0.57	0.86/0.16	0.55/0.70	0.51/0.71
	0.1	**0.20/0.92**	0.28/0.78	0.37/0.82	0.51/0.74	0.55/0.51	0.33/0.87	0.51/0.71
	0.15	**0.06/0.99**	0.06/0.96	0.25/0.90	0.29/0.89	0.37/0.70	0.20/0.93	0.51/0.71
Blurb (B)	0.02	**0.88/0.25**	1.00/0.00	0.80/0.35†	0.89/0.22	1.00/0.00	0.76/0.47†	0.66/0.56†
	0.05	**0.65/0.55**	0.92/0.10	0.67/0.55	0.76/0.47	0.98/0.02†	0.70/0.58†	0.66/0.56
	0.1	**0.44/0.75**	0.47/0.66	0.57/0.67	0.67/0.62	0.75/0.32	0.48/0.80†	0.66/0.56
	0.15	**0.27/0.90**	0.28/0.84	0.52/0.73	0.48/0.80	0.55/0.57	0.34/0.90	0.66/0.56
Amazon (A)	0.02	0.86/0.24	1.00/0.00	**0.74/0.44**	0.84/0.27	0.99/0.01†	0.68/0.54†	0.50/0.72†
	0.05	**0.49/0.67**	0.68/0.36	0.62/0.62	0.66/0.58	0.94/0.07	0.57/0.68	0.50/0.72
	0.1	**0.18/0.92**	0.19/0.87	0.46/0.75	0.55/0.74	0.53/0.53	0.29/0.89	0.50/0.72
	0.15	0.05/0.99	**0.03/0.99**	0.26/0.89	0.26/0.91	0.31/0.75	0.16/0.96	0.50/0.72

Validity. The realised error of HLCL exceeded α in at most 8% of splits in any setting, consistent with $\delta = 0.1$ (Figure 2); so did Item-cert. (10%) and Per-field (6%). HSC certifies the item-level error over all items, a different quantity, and when the annotator is weak its label-level error among accepted labels can be far above α: on WOS-46985 at $\alpha = 0.1$ it exceeded α in 100% of splits, because over a third of the items back off to the root, count as correct, and leave room for errors among the accepted labels. HSC is therefore treated as a valid competitor only in the rows where it met the label-level target. Depth-wise Benjamini–Hochberg exceeded α in up to 59% of splits although its mean realised error stays close to α (at most $1.05\,\alpha$): this is what an expected-value guarantee at the boundary looks like, not a failure of the method. The fixed threshold exceeded α in up to 100% of splits (on WOS-46985 it accepts labels with an error rate of 0.09), and item-level conformal selection in up to 24%.

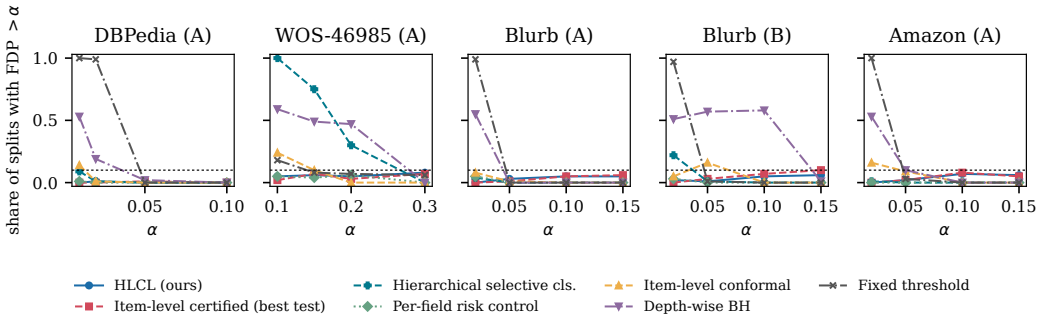


Figure 2: Share of splits whose realised label-level error exceeded α. The dashed line is $\delta = 0.1$.

Unit of decision. Item-cert. carries exactly the certificate of HLCL; we report its best test choice per setting, so the comparison isolates the unit of decision and, if anything, favours the baseline. At moderate risk, deciding per label accepts more correct labels: on the Blurb collection at $\alpha = 0.05$, 0.71 against 0.22 for annotator A and 0.55 against 0.10 for annotator B; on WOS-46985 at $\alpha = 0.2$, 0.61 against 0.31; and on the four-level Amazon taxonomy at $\alpha = 0.05$, 0.67 against 0.36. The gap shrinks as α grows (0.92 against 0.78 on the Blurb collection at $\alpha = 0.1$), and close to the annotator's own error rate both rules accept almost everything; the item-level rule can then be slightly cheaper because it leaves no item partially open (DBPedia at $\alpha = 0.05$: paired difference HLCL − Item-cert. $+0.019 \pm 0.045$).

Effort against the strongest certified baseline. In each setting we compare HLCL with whichever valid certified baseline has the lowest mean effort (Table 2). HLCL has the lower mean effort in 14 of the 20 settings. With the Nadeau–Bengio correction HLCL is significantly cheaper in 3 settings: on the Blurb collection against HSC at $\alpha = 0.05$ (-0.102 ± 0.068) and against Item-cert. at $\alpha = 0.1$ (-0.078 ± 0.063), and on Amazon against HSC at $\alpha = 0.05$ (-0.130 ± 0.054); there HLCL was cheaper in 99%, 99% and 100% of the splits. It is significantly more expensive in 1 setting, Amazon at $\alpha = 0.02$, against HSC ($+0.119 \pm 0.064$). In the remaining 16 settings the corrected intervals include zero. The largest of these are in favour of HLCL (WOS-46985 at $\alpha = 0.2$: -0.069 ± 0.097, cheaper in 89% of splits; at $\alpha = 0.3$: -0.042 ± 0.084), and the largest against it are DBPedia at $\alpha = 0.01$ ($+0.055 \pm 0.091$) and the Blurb collection at $\alpha = 0.02$ ($+0.069 \pm 0.082$), both against HSC. The correction is conservative; we report it because the splits overlap and a naive

interval would overstate significance.

Where HLCL *does not help.* Three regimes stand out. First, at the smallest α on DBPedia, the Blurb collection and Amazon, HSC is cheaper, significantly so on Amazon. There HLCL must use the depth-wise test, because the pooled test certifies little (on Amazon a non-empty rule in under half of the splits), and the depth-wise test asks every depth to meet α; HSC bounds the item-level error over all items, which is easier to meet when almost every item has a correct coarse label. Its mean label-level error stayed below α in these settings, although it exceeded α in 9% of splits on DBPedia. We cannot tell from these data whether the conservativeness of the depth-wise certificate or the label-level target itself is the main cause: the pooled test, which targets the same ratio, certifies little there either (effort 0.90 on Amazon). Second, when α is close to the annotator's error rate, the item-level rule with the same certificate is as cheap or cheaper (DBPedia at $\alpha = 0.05$). Third, on DBPedia at $\alpha = 0.02$ item-level conformal selection, which controls only an expectation, is the cheapest method among those that met the target (0.51 against 0.54); at $\alpha = 0.01$ it exceeded α in 14% of splits.

5.5. Ablations

Table 3 varies one component at a time under a fixed test.

Test statistic. The test has by far the largest effect. With the pooled test only, HLCL certifies almost nothing at the smallest α (effort 0.99 on DBPedia at $\alpha = 0.01$, 0.94 on the Blurb collection at $\alpha = 0.02$); with the depth-wise test only, it is much more expensive at larger α (0.53 against 0.20 on the Blurb collection at $\alpha = 0.1$). The selection rule picked the depth-wise test in all or nearly all splits at the smallest α of every corpus and in none at the two largest. It errs in the transition region: on DBPedia at $\alpha = 0.02$ it chose the depth-wise test in 26% of splits, and the automatic version needs 0.54 against 0.48 for the pooled test alone. The rule is evaluated on the fit split, which is half the size of the calibration split, so it is biased towards the depth-wise test near the transition; a rule that accounts for the calibration size would remove this bias.

Score, closure and per-depth thresholds. Under a fixed test these components matter in few settings. With the pooled test on DBPedia at $\alpha = 0.02$, the learned score and closure both help (effort 0.48 against 0.56 with the raw vote and 0.66 without closure), and the learned score saves a few points at

Table 3: Effort of HLCL variants (100 splits; lower is better; bold: within 0.005 of the best in the row and test group). Each variant is run with the automatic choice of test (left), with the pooled betting test only (middle) and with the depth-wise test only (right), so that the effect of a component is not confounded with the test that happens to be selected. *vote*: raw sample agreement instead of the learned score; *no cl.*: no ancestor closure, nodes accepted independently; *tied*: one threshold for all depths.

		automatic test				pooled test only				depth-wise test only			
Setting	α	full	vote	no cl.	tied	full	vote	no cl.	tied	full	vote	no cl.	tied
DBPedia (A)	0.01	0.82	**0.81**	0.82	1.00	**0.99**	1.00	1.00	1.00	0.82	**0.81**	0.82	1.00
	0.02	**0.54**	0.71	0.75	0.91	**0.48**	0.56	0.66	0.69	**0.75**	0.79	0.78	0.96
	0.05	0.13	0.14	0.14	**0.12**	0.13	0.14	0.14	**0.12**	0.32	0.35	0.32	**0.29**
	0.1	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	**0.00**	0.06	0.07	**0.05**	**0.05**
WOS-46985 (A)	0.1	**0.99**	**0.99**	**0.98**	1.00	**1.00**	**1.00**	**1.00**	**1.00**	**0.99**	**0.99**	**0.98**	1.00
	0.15	**0.85**	0.85	**0.85**	0.88	**0.85**	0.86	0.86	0.86	0.84	0.83	**0.82**	0.97
	0.2	0.63	0.66	0.63	**0.61**	0.63	0.66	0.63	**0.61**	0.73	0.76	**0.72**	0.76
	0.3	0.31	0.34	0.32	**0.30**	0.31	0.34	0.32	**0.30**	0.52	0.56	**0.50**	0.54
Blurb (A)	0.02	0.82	**0.82**	**0.82**	1.00	0.94	**0.94**	0.94	0.99	0.82	0.81	**0.81**	1.00
	0.05	**0.46**	**0.46**	0.47	**0.46**	**0.46**	**0.46**	0.47	**0.46**	0.67	**0.67**	**0.66**	0.99
	0.1	**0.20**	0.21	0.22	**0.20**	**0.20**	0.21	0.22	**0.20**	0.53	0.55	**0.51**	0.62
	0.15	**0.06**	**0.06**	0.07	0.06	**0.06**	**0.06**	0.07	0.06	0.34	0.35	**0.31**	**0.31**
Blurb (B)	0.02	**0.88**	**0.87**	**0.88**	1.00	0.99	**0.99**	**0.99**	0.99	**0.88**	**0.87**	**0.88**	1.00
	0.05	**0.65**	0.67	**0.65**	0.68	**0.65**	0.67	**0.65**	**0.66**	0.77	0.78	**0.76**	1.00
	0.1	**0.44**	**0.43**	0.46	0.46	**0.44**	**0.43**	0.46	0.46	**0.70**	**0.70**	**0.70**	0.97
	0.15	**0.27**	0.31	0.30	0.29	**0.27**	0.31	0.30	0.29	0.52	0.52	**0.47**	0.52
Amazon (A)	0.02	0.86	**0.84**	**0.85**	1.00	0.90	**0.81**	0.90	0.94	0.86	**0.84**	0.85	1.00
	0.05	0.49	**0.46**	0.50	0.51	0.49	**0.46**	0.50	0.51	0.68	**0.66**	**0.66**	1.00
	0.1	0.18	**0.17**	0.19	**0.17**	0.18	**0.17**	0.19	**0.17**	**0.59**	**0.59**	**0.59**	0.83
	0.15	**0.05**	**0.05**	0.06	0.05	**0.05**	**0.05**	0.06	0.05	0.33	0.32	**0.27**	0.28

larger α on WOS-46985 and with annotator B; on Amazon, by contrast, the raw vote is cheaper at every α and under every test, by up to nine points (0.81 against 0.90 at $\alpha = 0.02$). With the pooled test, removing closure never lowered effort by more than 0.005 in our 20 settings. With the depth-wise test it is the other way round: removing closure lowered effort by more than 0.005 in 13 of the 20 settings, by up to six points (Amazon at $\alpha = 0.15$: 0.27 against 0.33), because a depth-wise threshold no longer has to hold for nodes below a rejected parent. Per-depth thresholds matter with the depth-wise test, where a shared threshold must be strict enough for the hardest depth (e.g. 1.00 against 0.77 with annotator B at $\alpha = 0.05$, and 1.00 against 0.68 on Amazon at $\alpha = 0.05$); with the pooled test they mostly matter at small α (DBPedia at $\alpha = 0.02$: 0.69 against 0.48). We conclude that, in our data, the gains of HLCL come mainly from the label-level unit of decision, the pooled

Table 4: Sensitivity of HLCL to the review overhead c_0, the calibration size n_{cal} and the confidence level δ (Blurb collection, annotator A, 100 splits). The last three columns give the effort of the certified baselines under the same setting; Item-cert. here uses the automatic test selection, not the per-setting best variant of Table 2.

		HLCL				effort of baselines		
Setting	α	effort	power	fully auto	viol.	Item-cert.	HSC	Per-field
$c_0 = 0$	0.05	0.36	0.73	0.30	3%	0.91	0.43	0.50
$c_0 = 0$	0.1	0.15	0.92	0.69	5%	0.28	0.27	0.34
$c_0 = 0.5$ (default)	0.05	0.46	0.71	0.36	3%	0.91	0.56	0.67
$c_0 = 0.5$ (default)	0.1	0.20	0.92	0.71	5%	0.28	0.37	0.51
$c_0 = 1$	0.05	0.50	0.70	0.38	2%	0.91	0.63	0.75
$c_0 = 1$	0.1	0.22	0.91	0.71	5%	0.28	0.42	0.60
$n_{\text{cal}} = 300$	0.05	0.71	0.43	0.12	1%	1.00	0.59	0.72
$n_{\text{cal}} = 300$	0.1	0.28	0.87	0.59	0%	0.49	0.41	0.60
$n_{\text{cal}} = 600$	0.05	0.51	0.67	0.31	2%	0.99	0.58	0.70
$n_{\text{cal}} = 600$	0.1	0.24	0.89	0.65	3%	0.34	0.40	0.58
$n_{\text{cal}} = 1200$ (default)	0.05	0.46	0.71	0.36	3%	0.91	0.56	0.67
$n_{\text{cal}} = 1200$ (default)	0.1	0.20	0.92	0.71	5%	0.28	0.37	0.51
$\delta = 0.05$	0.05	0.47	0.70	0.35	2%	0.99	0.57	0.68
$\delta = 0.05$	0.1	0.22	0.91	0.68	1%	0.30	0.38	0.54
$\delta = 0.1$ (default)	0.05	0.46	0.71	0.36	3%	0.91	0.56	0.67
$\delta = 0.1$ (default)	0.1	0.20	0.92	0.71	5%	0.28	0.37	0.51
$\delta = 0.2$	0.05	0.44	0.73	0.39	6%	0.78	0.55	0.66
$\delta = 0.2$	0.1	0.19	0.93	0.72	7%	0.26	0.36	0.49

certificate and the choice between tests, and not from the learned score or from closure. Closure is nevertheless required for the output to be a valid path (Definition 1); in our experiments it saves effort under the pooled test, substantially only on DBPedia at $\alpha = 0.02$, and costs a few points under the depth-wise test.

Objective. Deploying the certified candidate with the most accepted labels instead of the cheapest one changes effort by at most a few points at $c_0 = 0.5$ (Table 3 does not show it; the largest change is 0.51 against 0.49 on Amazon at $\alpha = 0.05$). We did not run the label-maximising variant at other values of c_0.

5.6. Sensitivity

Table 4 varies one design parameter at a time on the Blurb collection.

Review overhead. The certificate does not depend on c_0; only the choice among certified thresholds does. A larger overhead pushes the deployed rule towards accepting whole items: at $\alpha = 0.05$ the share of fully automatic items

rises from 0.30 at $c_0 = 0$ to 0.38 at $c_0 = 1$, while power changes from 0.73 to 0.70. The violation share is unchanged, and HLCL remains the cheapest certified method for every c_0.

Calibration size. With fewer calibration items the certificate becomes more conservative, as expected: at $\alpha = 0.05$ effort rises from 0.46 with 1,200 items to 0.51 with 600 and 0.71 with 300. At $\alpha = 0.1$ the item-level rule (automatic variant) loses more: its effort rises from 0.28 to 0.49, against 0.20 to 0.28 for HLCL. With only 300 calibration items and $\alpha = 0.05$, HSC is cheaper than HLCL; a plausible reason, which we have not verified, is that its single binomial certificate on the item-level error needs less data than HLCL's pooled label-level one. In these runs the selection rule always chose the pooled test, because it looks only at the fit split and cannot react to the calibration size. The violation share stays small throughout.

Confidence level. Changing δ from 0.05 to 0.2 moves effort by about two points at both values of α, and the violation share tracks δ as it should (1%, 5% and 7% of splits at $\alpha = 0.1$).

5.7. Second annotator

Annotator B is less accurate than annotator A on the Blurb collection at every depth (Table 1). The guarantee still holds at every α, and HLCL has the lowest mean effort among the certified methods at all four values of α, by one to three points; none of these differences is significant after the Nadeau–Bengio correction (Table 2). The score and the procedure were applied to annotator B without any change.

6. Discussion

Which test when. The pooled betting test should be the default: it uses the pooled constraint that operators care about and is sharper when per-depth error rates differ. The depth-wise test is the fallback for small α, where the pooled test certifies little or nothing. The selection rule makes the switch automatically from the fit split; the chosen test is recorded in every released result file. Because the rule does not look at the calibration size, it does not adapt to small calibration splits (§5.6).

High-probability or expected guarantee. The high-probability guarantee and the expected-FDP guarantee of conformal labeling bound different functionals, and neither was uniformly cheaper in our experiments: conformal labeling was slightly cheaper on DBPedia at $\alpha = 0.02$ but more expensive at the smallest α on the Blurb collection, and where it was much cheaper its realised error often exceeded the target. An operator who re-calibrates often and cares about average quality over many deployments may prefer the expected form; an operator who must defend the quality of one released dataset needs the high-probability form. Both can share the node score and the ancestor-closed rule.

Practical use. The review step need not be a human; it can be a stronger model, a rule engine or a domain expert, and the weights ω_d can reflect the cost of each level. In a data platform the outcome can be stated as a column policy, i.e. which levels of which taxonomy are written without review at which certified error, and audited later on a fresh labelled sample.

Limitations. The guarantee assumes that calibration and deployment items come from the same distribution; drift in the topic mix invalidates it, as for any calibration-based method. Items that share a source, such as several messages of one thread, are not independent; the procedure should then be applied at the level of the independent unit, as discussed in [6]. We assume a fixed tree taxonomy with one gold path per item; multiple gold paths and DAG taxonomies need a different notion of correctness. The annotator is queried five times per item, and we have not studied how the score degrades with fewer samples. The effort model is a stylised default (c_0, ω_d) rather than a measured review time; we show sensitivity to c_0 but have not run a timed study. Each corpus contributes 3,000 items, the deepest taxonomy has four levels, and DBPedia is easy at the coarse levels, so the DBPedia results mainly test the small-α regime. Finally, the test-selection rule is a heuristic: it is valid by construction but not optimal.

7. Conclusion

We studied label-level selective acceptance of hierarchical LLM annotations under a certified error bound and an explicit reviewing cost. HLCL decides per node, keeps accepted labels ancestor-closed, certifies per-depth thresholds with a learn-then-test procedure whose design choices are all fixed

on held-out data, and deploys the cheapest certified threshold. On four public corpora of depth two to four with two annotators, the share of splits whose realised error exceeded the target stayed below $\delta = 0.1$ (at most 8%). The evidence is limited to 3,000 items per corpus, taxonomies of depth at most four and one second annotator. Under the same certificate, deciding per label rather than per item accepted substantially more correct labels at moderate risk. Against the strongest certified baseline in each setting, HLCL had lower mean effort in most settings, but after correcting for overlapping splits it is significantly cheaper in 3 and significantly more expensive in 1; at very small α hierarchical selective classification, which certifies a different quantity, was cheaper, and close to the annotator's error rate the item-level rule was as good. The ablations suggest that the gains come from the unit of decision and the certificate rather than from the learned score. Code, cached annotations and per-split results are released.

Data and code availability

The four corpora are public: DBPedia classes (licence CC0; Hugging Face dataset `DeveloperOats/DBPedia_Classes`), WOS-46985 (Mendeley Data, dataset `9rw3vkcfy4`), the Blurb Genre Collection (Universität Hamburg, licence CC BY-NC 4.0) and the Home & Kitchen metadata of Amazon Reviews 2023 (Hugging Face dataset `McAuley-Lab/Amazon-Reviews-2023`). We redistribute no text from any corpus; users obtain the corpora from their sources under the providers' terms. The code, the cached LLM responses (labels and stated confidences only, no source texts), the per-split result files and the scripts that regenerate every table and figure are available at `https://github.com/wanghongsheng01/hlcl` (archived with a DOI on Zenodo at acceptance).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI coding and writing assistant to implement the experiments, to check references and to draft and edit the text. All results were produced by running the released code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Hongsheng Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Fei Wu**: Supervision, Writing – review & editing.

Declaration of competing interest

Hongsheng Wang is affiliated with Loctek Ergonomic Technology Corp., whose e-mail annotation workload motivated this study. The company had no role in the design of the method, the choice of datasets, the analysis or the decision to publish, and no company data are used. The authors declare no other competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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Appendix A. Proofs

Ancestor closure of (5). If $A_{i,d} = 1$ then $\mathbf{1}\{s_{i,e} \geq \lambda_e\} = 1$ for every $e \leq d$, and hence $A_{i,e} = 1$ for every $e \leq d$.

Validity of the depth-wise p-value. Fix λ and d. The pairs $(A_{i,d}, W_{i,d})$ are i.i.d. across items, with $W_{i,d} \le A_{i,d}$. Conditionally on $(A_{1,d}, \dots, A_{n,d})$, the variables $W_{i,d}$ of the items with $A_{i,d} = 1$ are independent Bernoulli with parameter $\Pr(W_{i,d} = 1 \mid A_{i,d} = 1) = q_d$. Hence $W_d = \sum_i W_{i,d} \mid n_d \sim \mathrm{Bin}(n_d, q_d)$, and under $H_d : q_d > \alpha$ the binomial tail (9) satisfies $\Pr(p_d \le t \mid n_d) \le t$ for all t, because the $\mathrm{Bin}(n_d, q)$ distribution is stochastically increasing in q. Integrating over n_d gives $\Pr(p_d \le t) \le t$. If $n_d = 0$ and $\lambda_d < \infty$ we set $p_d = 1$, which is trivially valid. If $\lambda_d = \infty$ the rule cannot accept at depth d, so $\mathbb{E}[A_d] = 0$, depth d does not enter (8), and it is omitted from the maximum. By (8), $\mathrm{FDR}(\lambda) > \alpha$ implies that H_d holds for some d, say d^*. Then $\Pr(\max_d p_d \le t) \le \Pr(p_{d^*} \le t) \le t$, so p^{iut} is valid for H_λ.

Proof of Theorem 1. Condition on the fit split. The candidates $\lambda_{(1)}, \lambda_{(2)}, \dots$, their order and the test statistic are then fixed, and the calibration items are i.i.d. and independent of them, so each $p_{(j)}$ is a valid p-value for $H_{\lambda_{(j)}}$. Fixed-sequence testing rejects $H_{\lambda_{(1)}}, \dots, H_{\lambda_{(J)}}$ where J is the index before the first p-value above δ. Let j^* be the smallest index whose null is true. A true null is rejected only if $J \ge j^*$, which requires $p_{(j^*)} \le \delta$; hence the probability of rejecting any true null is at most $\Pr(p_{(j^*)} \le \delta) \le \delta$ [19]. On the complementary event, which has probability at least $1-\delta$, every certified λ has $\mathrm{FDR}(\lambda) \le \alpha$ by (7) (a candidate with $\mathbb{E}[A(\lambda)] = 0$ accepts nothing and is harmless). The deployed $\hat{\lambda}$ is one of the certified candidates or accepts nothing, which proves (2). The argument is conditional on the fit split and therefore also holds unconditionally.